ARTIFICIAL INTELLIGENCE · BCSE306L

TWINFORGE 3D

AI-POWERED GENERATION OF DIGITAL TWINS FROM PHYSICAL ROBOTS

TEAM

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REGISTRATION NUMBER

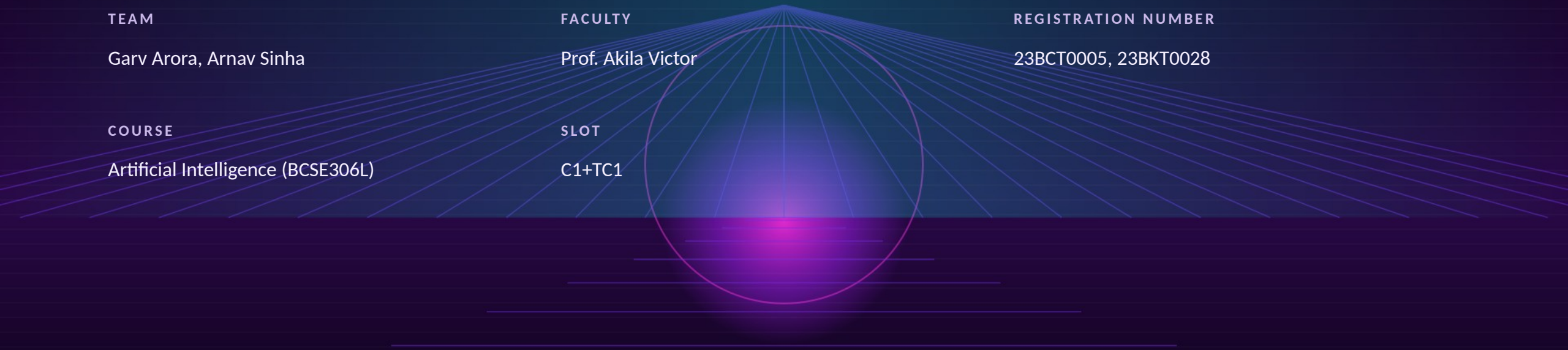
23BCT0005, 23BKT0028

COURSE

Artificial Intelligence (BCSE306L)

SLOT

C1+TC1



Physical Robot → 3D Scan → AI Understanding → Kinematic Reconstruction → URDF → Simulation

Project Aim

To design and develop TwinForge 3D — an AI-powered system that automatically converts a real-world robot into a simulation-ready digital twin (URDF) by reconstructing its geometry, understanding its kinematic structure, and generating a validated robot description without manual CAD or URDF authoring.

1

Automate reconstruction of a robot's 3D geometry and component structure

2

Infer joints, joint types, and kinematic relationships using AI + geometry

3

Generate a complete, simulation-ready URDF automatically

4

Validate the digital twin against the physical robot's real motion

Abstract

Physical Robot → 3D Scan → AI Understanding → Kinematic Reconstruction → URDF → Digital Twin

Creating a digital twin of a physical robot is traditionally a slow, manual process involving measurement, CAD modelling, joint identification, and hand-written URDF files — a workflow that can take hours or days even for moderately complex robots. TwinForge 3D automates this pipeline end-to-end. Using RGB, RGB-D, LiDAR, or smartphone scans, the system reconstructs the robot's 3D geometry, segments it into meaningful components, and applies AI-based reasoning combined with classical geometric algorithms to identify links, joints, joint types, and kinematic axes. Where a single static scan is insufficient, the system leverages multiple robot poses and an active-scanning loop to resolve uncertainty. The result is a validated, simulation-ready URDF that can be imported directly into Gazebo, RViz, Isaac Sim, or Webots — collapsing a multi-day manual workflow into a guided, largely automatic one.

15

Sub-objectives automated

4+

Simulator targets (Gazebo, RViz, Isaac Sim, Webots)

5

Robot classes supported

Existing Solutions & Their Limits

1 Manual URDF Authoring (ROS / xacro)

Engineers hand-measure links, write XML, and iterate in a simulator.

LIMITATION

Extremely time-consuming; error-prone; not scalable

2 CAD-to-URDF Plugins

Tools (e.g. SolidWorks-to-URDF) export kinematics from existing CAD assemblies.

LIMITATION

Requires a pre-existing accurate CAD model of the robot

3 Photogrammetry / 3D Scanning Apps

Apps like Polycam or RealityCapture reconstruct visual 3D geometry.

LIMITATION

Produce only static meshes — no joints, no kinematics

4 Research: Articulated-Object Reconstruction

Recent AI models (e.g. URDF-Anything, Articulate-Anything) infer articulation from images.

LIMITATION

Focused on generic household objects, not full robot systems — reviewed next

Papers 1 – 3

PAPER 1

URDF-Anything

arXiv:2511.00940

A 3D multimodal LLM that takes point-cloud + text and predicts part segmentation with joint parameters; SOTA on PartNet-Mobility.

GAP

Evaluated on household objects, not robot hardware; needs offline MLLM training.

HELPS US

Confirms one AI model can jointly reason over geometry and kinematics.

WE FILL

We extend to real robots with active multi-view scanning & multi-sim export.

PAPER 2

Articulate-Anything

arXiv:2410.13882

An agentic vision-language pipeline generating URDF-compilable articulation code, with a render-and-critique self-correcting loop.

GAP

Relies on retrieving parts from mesh libraries (ShapeNet/PartNet).

HELPS US

Its render-critique loop inspires our confidence-driven refinement loop.

WE FILL

We reconstruct true 3D geometry from the user's own scan instead of retrieval.

PAPER 3

Articulat3D

arXiv:2603.11606

Reconstructs articulated digital twins from monocular smartphone video using geometric and motion constraints.

GAP

Assumes independent joints, not a hierarchical kinematic tree.

HELPS US

Validates our Motion-Based Kinematic Reconstruction layer.

WE FILL

We generalize to full hierarchical trees (base → links → joints).

Papers 4 – 5

PAPER 4

ScrewSplat

arXiv:2508.02146

Infers kinematic structure by splatting screw axes with 3D Gaussians from multi-view RGB, generalizing to unseen categories.

GAP IN THIS PAPER

Outputs kinematics only — no URDF, no physical parameters, no guidance for uncertain views.

HOW IT HELPS US

Its category-agnostic multi-view screw-axis estimation is a strong candidate for our Joint Axis Estimation module.

WHAT WE FILL

We wrap axis estimation inside a full pipeline adding physical parameters, active rescanning, and validated URDF export.

PAPER 5

LARM

arXiv:2511.11563

Reconstructs part meshes and estimates joint axes (revolute/prismatic) from six images per part, classifying joint type at 88.89% accuracy.

GAP IN THIS PAPER

Treats joint type as a given input; not tailored to full robot systems (sensors, end-effectors, multi-link chains).

HOW IT HELPS US

Shows per-part multi-view capture + VLM-assisted joint classification is practical and accurate.

WHAT WE FILL

We infer joint type automatically and target complete robot classes with direct multi-simulator integration.

Gaps Across Existing Approaches

1	**No robot-specific end-to-end pipeline**	Prior work targets household articulated objects (cabinets, doors), not full robot classes — arms, rovers, legged, aerial.
2	**No active / guided rescanning**	Systems rarely detect low-confidence joints and request an additional, targeted observation from the user.
3	**Missing physical parameter estimation**	Mass, center of mass, and inertia are usually left out, even though simulators need them for physics-accurate twins.
4	**No closed-loop real-vs-sim validation**	Few methods compare the digital twin's simulated motion back against the physical robot to refine parameters.
5	**Limited multi-simulator export**	Outputs are rarely packaged as directly importable, validated URDFs for Gazebo, RViz, Isaac Sim, and Webots.

TwinForge 3D — Not Just a Mesh, a Kinematic Tree

Instead of producing just a static 3D mesh, TwinForge 3D reconstructs the robot as a robotic system — a kinematic tree of links and joints — and converts that understanding directly into a validated URDF.

SYSTEM CAPABILITIES

Multi-modal capture

RGB, RGB-D, LiDAR, or smartphone video, from multiple viewpoints

AI + geometry hybrid

Foundation vision models for semantics; classical ICP / registration for precise transforms

Motion-based reasoning

Compares multiple robot poses to infer real joint axes and types

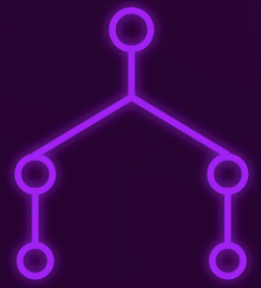
Active scanning loop

Flags low-confidence joints and requests a targeted follow-up scan

Simulator-ready output

Validated URDF exportable to Gazebo, RViz, Isaac Sim, and Webots

A ROBOT KINEMATIC TREE



```
base_link
  └ joint_1 (revolute)
      └ link_1
          └ joint_2
              └ link_2
                  └ end_effector
```

Methodology

01

Data Acquisition

Capture the robot via RGB, RGB-D, LiDAR, or smartphone across multiple viewpoints and, where possible, multiple poses.

02

3D Reconstruction

Build a point cloud / mesh using structure-from-motion, RGB-D fusion, or SLAM; clean and segment it.

03

AI Perception & Segmentation

Vision foundation models identify components, the base, and which parts likely move relative to each other.

04

Joint Detection & Axis Estimation

Classical geometric algorithms (ICP, registration) plus AI reasoning determine joint type, axis, and confidence.

05

Kinematic Tree Construction

Parent-child relationships and coordinate frames are assembled into a complete kinematic hierarchy.

06

Parameter Estimation & URDF Generation

Mass, COM, and inertia are estimated; visual/collision meshes are generated; a full URDF is written automatically.

07

Validation & Refinement

The digital twin's simulated motion is compared with the physical robot; low-confidence parts trigger a rescan.

High-Level System Architecture



DIGITAL TWIN

Gazebo

RViz

Isaac Sim

Webots

SUPPORTED ROBOT TYPES

Arms · Rovers · Legged · Aerial · Custom

Deploys directly to industry-standard simulators from a single validated URDF export.

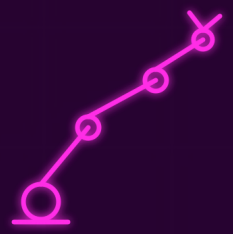
Primary objective & the fifteen supporting milestones

PRIMARY OBJECTIVE

Automatically generate a simulation-ready digital twin of a physical robot from 3D observations.

1 Reconstruct 3D geometry	**2** Identify components	**3** Separate links	**4** Identify joints	**5** Determine joint types
6 Estimate joint axes	**7** Map parent-child links	**8** Build kinematic tree	**9** Generate coordinate frames	**10** Generate visual/collision geometry
11 Estimate physical parameters	**12** Generate complete URDF	**13** Validate generated model	**14** Refine with new observations	**15** Support diverse robot types

Built for every kinematic form factor



Robotic Arms

4/6-DOF, collaborative & educational arms



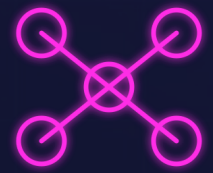
Mobile Robots

Differential-drive, 4-wheel, tracked, omni



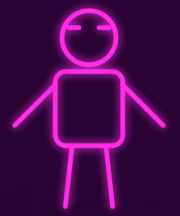
Legged Robots

Quadrupeds, hexapods, humanoid-style



Aerial Robots

Drones, quadcopters, multi-rotor platforms



Other Platforms

Mobile manipulators, lab & research robots

Data acquisition & 3D reconstruction

LAYER 1 — DATA ACQUISITION

RGB Camera

Color, appearance, component recognition

RGB-D Camera

RGB + depth + spatial information

LiDAR

Highly accurate spatial geometry & point clouds

Smartphone

RGB, depth where available, multi-view video

LAYER 2 — 3D RECONSTRUCTION

Point Cloud

↓

Cleaned Point Cloud

↓

Segmented Point Cloud

↓

3D Mesh

Techniques: SfM, multi-view stereo, RGB-D reconstruction, SLAM, neural 3D reconstruction, registration

AI-based robot perception & component segmentation

QUESTIONS THE AI MUST ANSWER

- *What components exist?*
- *Which geometry belongs to which component?*
- *Which component is the base?*
- *Which components appear connected?*
- *Which are likely to move relative to one another?*

COMPONENT SEGMENTATION

Rover

- Chassis
- Wheel 1
- Wheel 2
- Wheel 3
- Wheel 4



Each detected component becomes a candidate URDF link.

Detecting joints and reasoning about motion



JOINT TYPES

Fixed · Revolute · Continuous
Prismatic · Floating · Planar

MOTION-BASED KINEMATIC RECONSTRUCTION

By comparing geometry across multiple robot poses, the system infers:

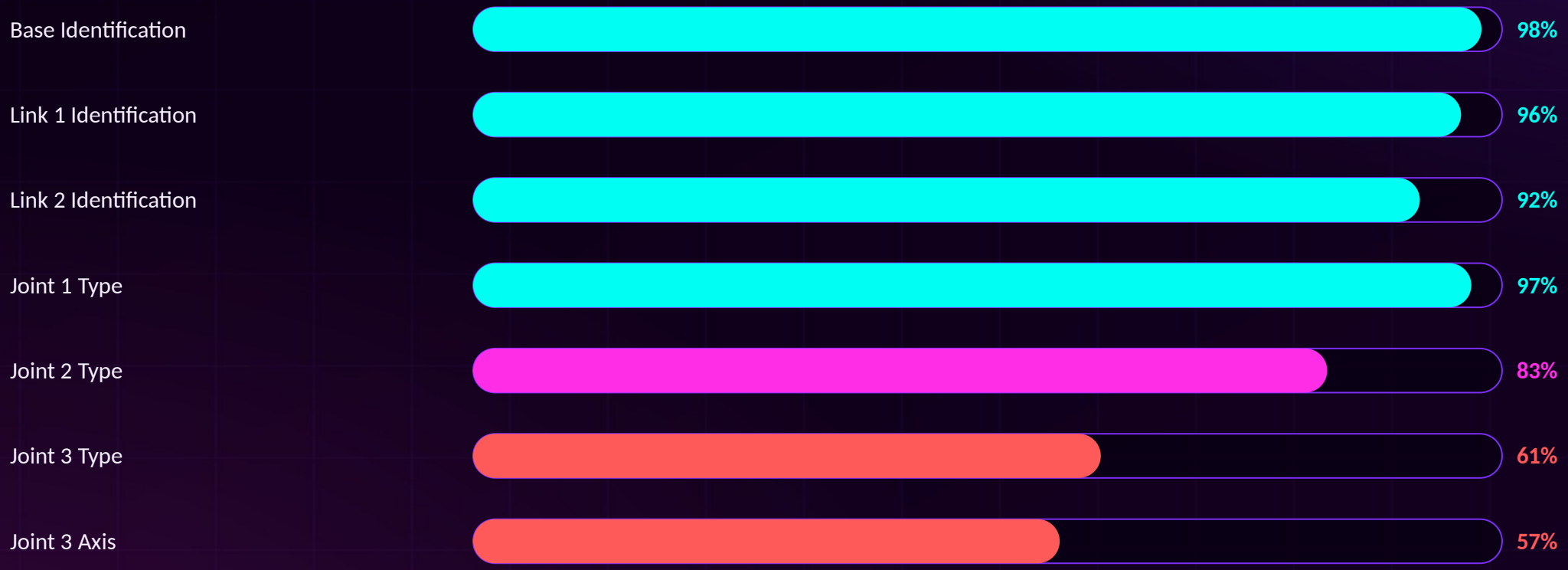
- Which parts moved
- Relative transformations
- Rotational & translational motion
- Likely joint axes

ACTIVE SCANNING LOOP

- Initial Scan
- Uncertainty Detection
- Required Observation
- User Moves Robot
- New Scan
- Improved Model

Every inference carries a confidence score

Low-confidence elements trigger a request for additional observations — the system never blindly generates a model from uncertain information.



Joint 3 has low confidence — rotate the arm ~20° and scan again.

A complete, validated robot description — auto-generated



URDF STRUCTURE

```
robot
├── links
│    ├── base_link
│    ├── link_1 / link_2
│    └── end_effector
├── joints
│    ├── joint_1 / 2 / 3
├── visual & collision models
├── inertial parameters
└── transformations
```



VALIDATION CHECKS

Structural

Joints have parents/children; no invalid cycles

Geometric

Meshes load correctly; transforms are valid

Kinematic

Joint axes and hierarchy are consistent

Simulation

Load into simulator; compare expected motion

AI + geometry + robotics — not one model doing everything

Computer Vision

SAM 2 · YOLO · DINO · VLMs

3D Processing

Open3D · PointNet++ · NeRF/Gaussian · TSDF

AI Reasoning

VLMs · GNNs · transformer 3D models

Geometric Algorithms

ICP · registration · axis fitting · optimization

Frontend

React · Three.js · React Three Fiber

Backend

Python · FastAPI

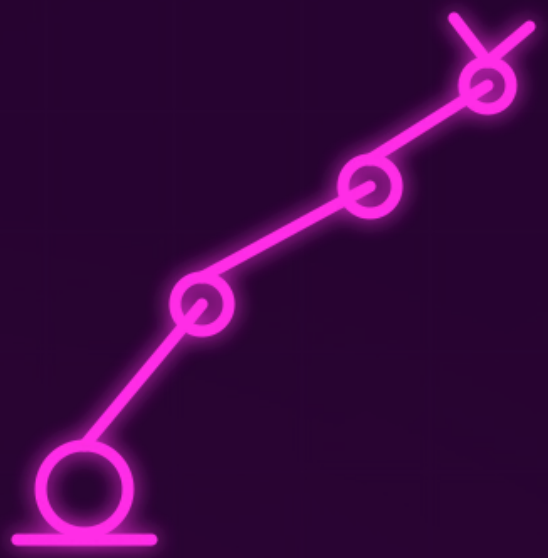
Robotics

ROS 2 · URDF · TF2 · MoveIt 2

Simulation

Gazebo · Isaac Sim · Webots

A 6-DOF robotic arm, end to end



Physical 6-DOF Robot Arm — scanned from multiple angles and poses.

DETECTED STRUCTURE

- ✓ Base
- ✓ Link 1
- ✓ Link 2
- ✓ Link 3
- ✓ Link 4
- ✓ Link 5
- ✓ End Effector

6 Revolute Joints Identified

Links: 7 · Joints: 6 · Confidence: 93% · URDF: Generated · Validation: Passed

References

[1] Y. Liu et al., "URDF-Anything: Constructing Articulated Objects with a 3D Multimodal Large Language Model," arXiv:2511.00940, 2025.

[2] L. Le et al., "Articulate-Anything: Automatic Modeling of Articulated Objects via a Vision-Language Foundation Model," arXiv:2410.13882, 2024.

[3] "Articulat3D: Reconstructing Articulated Digital Twins from Monocular Videos with Geometric and Motion Constraints," arXiv:2603.11606, 2026.

[4] "ScrewSplat: An End-to-End Method for Articulated Object Recognition," arXiv:2508.02146, 2025.

[5] "LARM: A Large Articulated-Object Reconstruction Model," arXiv:2511.11563, 2025.

[6] Open3D — "A Modern Library for 3D Data Processing," Open3D Documentation, open3d.org.

[7] Meta AI Research, "Segment Anything (SAM)" and the SAM 2 follow-up work — referenced as vision-foundation segmentation models.

[8] Open Robotics, "URDF — Unified Robot Description Format" and "ROS 2 Documentation," ros.org.

Papers 1–5 (used for the literature review) are freely available on arXiv.

THANK YOU

TwinForge 3D · AI-Powered Generation of Digital Twins from Physical Robots

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